Electrical Impedance Tomography Deep Imaging With Dual-Branch U-Net Based on Deformable Convolution and Hyper-Convolution

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***Abstract*— Electrical impedance tomography (EIT) is a new imaging modality for nonionizing radiation that uses a safe current applied to the surface and measures the response voltage on a boundary sensor to solve an inverse problem of determining the conductivity distribution in the region of interest (ROI). Due to the "soft-field" nature of the electrical field, the boundaries of the inclusions are usually blurred and the conductivity parameters are inaccurate in the reconstructions. To address the above problems, this article proposes a learning-based two-branch U-Net deep imaging architecture, named DHU-Net, for the accurate and sharp reconstruction of EIT images. Specifically, deformable convolution layers are introduced to improve the representation of shape and spatial information by a small convolutional kernel, while SE-Attention is used to recalibrate the channel-wise features for global distributions; on the other hand, an implicit hyper-convolutional network with coordinate attention (CA) is used to construct the relationship between the spatial coordinates of the convolutional kernel and corresponding weights, so that the large convolutional kernel for conductivity recovery has a relatively lower parameter while having better convolutional robustness. DHU-Net is trained and fine-tuned using a large number of simulation samples and validated on tank experiments. The metrics show that the RMSE is 2.0637, the SSIM is 0.9522, and the RSNR is 46.4597 (the metrics are lower than those of the TR method by 68.76%, 56.53%, and 61.00%) which provides better reconstruction visualization and consistency between the quantification metrics and reconstruction results compared to the state-of-the-art models. The experimental results show that the DHU-Net proposed in this article has better robustness and reconstruction consistency, suggesting that deformable convolution and implicit neural networks have better expressive capabilities in shape modeling, as well as less computational costs and faster inference than dense-based models, which can drive real-time applications of EIT for structural and functional imaging.**

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***Index Terms*— Attention methods, deformable convolution, electrical impedance tomography (EIT), hyper-convolution, U-Net.**

I. Introduction

ELECTRICAL impedance tomography (EIT) is a noninvasive, real-time, low-cost, and emerging imaging technique used to analyze electrical distributions in a region of interest (ROI). It relies on measuring how an object modulates excitation currents and employs inverse problem computational methods to reconstruct object distributions [1]. EIT finds applications in various fields, including industrial process monitoring [2], biomedical imaging (e.g., lung respiration monitoring, brain stroke therapy) [3], [4], and more. However, the EIT inverse problem faces challenges like nonlinearity, ill-posedness, and prone to artifacts due to sensor movement and complex contact impedance [5]. To address these issues, regularization-based and optimized iterative methods, such as hybrid frameworks (Lp-norm, total variation, wavelet basis function) and relaxation Gauss-Newton, are employed [6], [7]. Nonetheless, these numerical reconstruction methods (optimized Land-weber iteration, FISTA, back projection, and D-Bar, group sparse-based method, etc. [8], [9], [10], [11], [12], [13], [14]) often rely heavily on prior models and empirical criterion, have high computational complexity, and limited applicability across scenarios.

A. Prior Works: Learning-Based Methods for Electrical Tomography(ET)

In recent years, deep learning methods have gained greater success in the field of computer vision and have been used in inverse problem imaging [15], [16] to achieve tasks such as medical image reconstruction [17], super-resolution reconstruction [18], image denoising [19], etc. EIT imaging, as a typical nonlinear inverse problem, has also been used by a wide range of researchers to achieve high-quality image reconstruction using deep learning-based models [20]. Currently, such methods can be divided into two categories. The first idea is the direct reconstruction of the manifold transformation relationship between the boundary measurement and the image of the electrical distribution, which is called "data-driven" imaging. For example, Chen et al. [21] proposed a multilayer autoencoder (MLAE) feature extraction-image reconstruction model; Fu et al. [22] proposed a feature encoding- decoding parallel subnetwork method based on $L2$-norm regularization guided deep EIT imaging (RGDI) and incorporating spatial location of the sensor distribution to increase a priori knowledge, Seo et al. [23] mathematically analyzed the theory of variational networks to the boundary problem and proposed a model for reconstructing the EIT image of the lungs based on the manifold prior; Herzberg et al. [24] first used graph convolutional network (GCN) for EIT image reconstruction, realized the unity between the forward problem mesh and the specific inverse problem grid of finite element calculation, Liu et al. used the deep image prior (DIP) strategy to design an untrained deep learning model to generate EIT images with clear boundaries from random noise [25]. The other idea is to use numerical calculations or initial imaging methods to obtain "coarse" distributions, and then deep learning methods are utilized as a post-processing network to enhance the boundary features and electrical parameters, which is known as an "image-driven" imaging method. For example, Hamilton and Hauptmann [26] proposed the deep D-Bar algorithm, in which the initial image obtained by the truncated low-pass D-Bar method is used as the U-Net input, and a sharper EIT image of the lungs is obtained by a supervised training approach. Zhang et al. [27] proposed that enhanced WGAN improves EIT image reconstruction using a fully connected network to establish nonlinear mappings between measurements and image distribution. The full connection restricts spatial resolution to 32×32 due to model parameters, moreover, the fixed kernel in WGAN is not suitable to express the complex boundaries [27]. Ren et al. [28] use manifold theory to establish the Tikhonov regularization (TR) as a neural model to obtain initial pixel weights and a two-scale residual network is established to construct correspondences between pixels to improve the robustness of the shape reconstruction. Wang et al. [29] propose that V^2A-Net uses a channel-wise and coordinate-wise attention parallelism strategy to optimize the skip connection of the symmetric encoding-decoding structure, which allows for the simultaneous completion of high-resolution EIT images and shape reconstruction tasks. It can be seen that the spatial resolution of the "data-driven" imaging method is often lower than that of the "image-driven" method, which is mainly due to the limitation of the number of neural of the network, and in addition, the current EIT deep imaging method often adopts the regular convolution kernel to deal with the spatial relationship of pixel matrices but ignores the information of multiple shape features of the inclusions, which is reflected in the reconstructed images as a result that the curvature and the sharpness of the boundaries cannot be accurately expressed, even though artifacts are obviously suppressed.

B. Challenges and Motivations

Regular convolutional neural networks (CNNs) inherently limit the ability to model geometric deformation due to the fixed kernel in their structure, resulting in poor performance in complicated shape representation. Deformable ConvNets (DCNs) have revolutionized the representation of complex geometric features in various image-related tasks. In, Dai et al. [30] introduced DCN, which replaces traditional square convolution kernels, enhancing geometric modeling capabilities. In 2018, they extended DCN with an image energy modulation operator to distinguish ROI from irrelevant information in deeper DCN layers [31]. Chen et al. [32] further improved image segmentation with adaptive deformable convolutions using multiple attention mechanisms. Integrating DCNs with feature extraction has also yielded impressive results. Ji and Tekalp [33] applied a deformable convolutional residual module to super-resolution tasks, Yang et al. [34] used a residual-connected DCN network for blood vessel segmentation, and Li et al. [35] developed a multitasking framework for geometric deformations. DCNs have found applications in visual inverse problems, such as 4D-CBCT reconstruction [36] and cardiac MRI imaging [37], Shen et al. [38] achieved multiorgan CT segmentation using DCNs, while Zuo et al. [39] introduced a deformable attention module for visual recognition tasks, Wang et al. [40] proposed an angular deformable alignment module for super-resolution of light field images. These approaches employ deformable convolutions to enhance deep learning models in modeling geometric shapes across a range of image tasks.

Another essential problem with typical convolution is that the expressive performance of their models is directly related to the kernel weights of the convolution kernel (i.e., the number of channels, the scale of the convolution kernel), and in general, the larger the convolution kernels the more global field of view it can capture, but at the same time, it requires more computational resources to be consumed. Alternative convolutional kernels are widely used such as depth-wise separable convolutional filters [41], Dilated convolutions [42], [43], and atrous spatial pyramid pooling [44], [45] can be effective in reducing the model parameters, but they maintain a strong coupling between the model capacity and the number of pixels used in the kernel. This coupling can lead to over-fitting or limit the expressiveness of various tasks that require capturing long-range dependencies and generating dense pixel-level predictions. Moreover, nonlocal networks [46], self-attention (SA), and Transformer-based framework [47] have gained widespread attention in computer vision tasks due to their ability to aggregate long-range dependencies, but require more expensive computational resources. Hyper-Networks, which have the advantage of generating weights for another network responsible for the main task, can improve parameter efficiency and provide flexibility without significantly sacrificing expressiveness, gaining attention on image segmentation, reconstruction tasks, such as HyperSeg for CT image segmentation [48], HyperRecon for MRI image reconstruction [49]. Alternatively, neural networks have been used to create implicit representations of signals allowing convolutional kernels to be constructed without being limited to a fixed size, and model weights can be constructed through implicit learning, which is used for related tasks such as cardiac MRI reconstruction [50], [51].

C. Contributions

Aiming at the poor performance of shape modeling performance caused by the use of fixed convolutional kernels and

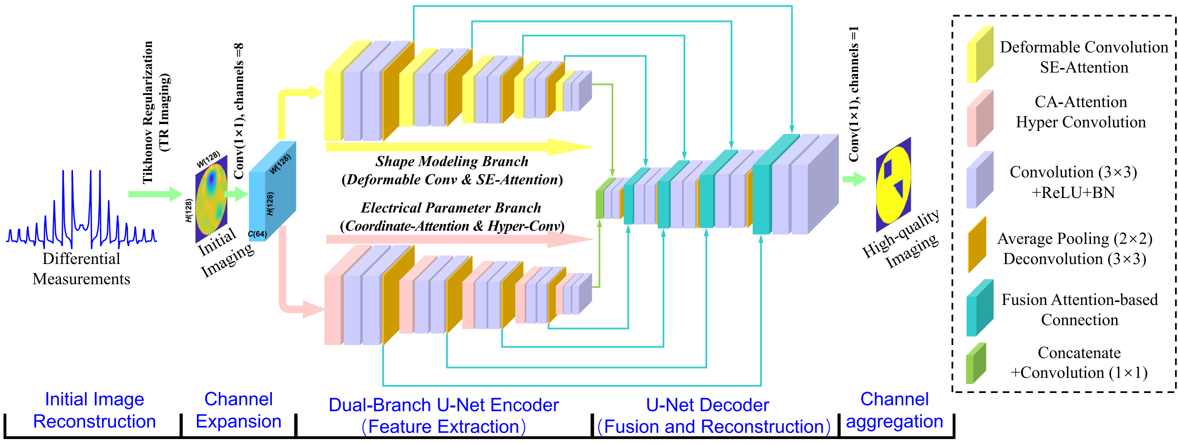


Fig. 1. Overview of DHU-Net architecture, which contains some processing stages, such as 1×1 convolution (with channels $= 8$) for channel expansion, feature extraction, image reconstruction, and channel aggregation by 1×1 convolution (with channel $= 1$) for image presentation.

the large number of parameters caused by complex feature connections in the current EIT imaging models based on deep learning-based architectures, this article proposes a two-branch EIT deep imaging model, in which the U-Net is the backbone with DCNs and hyper-convolutional networks, is referred to DHU-Net. The main contributions of DHU-Net are listed as follows.

1) The "image-driven" idea is adopted in DHU-Net. The U-Net is the backbone of the deep imaging model, where an improved two-branch encoder with different kernels for feature extraction and a fusion decoder for image reconstruction is used for the electrical parameter reconstruction and shape reconstruction tasks in EIT imaging, respectively.
2) The DCN is introduced to implement shape feature extraction, and SE-channel attention also utilizes to refine the degree of attention to different degrees of shape in DCN. This design can better focus on ROI shape characteristics and avoid the influence of background parameters.
3) To reduce the traditional parametric representation using large convolutional kernels for global feature learning, implicitly expressed HyperConvNets are used to optimize the number of model parameters, while orientation feature maps are extracted through coordinate attention (CA) prior to the input features, which in turn enables implicit convolutional kernel weight learning.
4) A novel skip connection between the encoder and decoder is designed carefully. In order to fully reuse the global and local information, skip-attention connections based on SA are introduced in the DHU-Net, which can be better to aggerate the low-level and high-level features than single connections.

DHU-Net was trained and parameter fine-tuned using a large number of simulation samples, and image reconstruction is performed under real-world visualization experiments, and the results show that the method has better noise robustness and model generalization ability. The article is structured as follows: a brief description of the EIT sensing principle and mathematical model is given in Section II, the DHU-Net model structure and design details are given in Section III, the relevant experimental setup as well as the evaluation metrics are given in Section IV, the reconstruction results and quantitative values are analyzed and compared in Section V, and some discussions on DHU-Net are given in Section VI, and finally the corresponding conclusions are given.

II. METHODOLOGY

Considering the ill-posed and nonlinear nature of EIT, deep learning-based methods with the help of powerful nonlinear representations are considered a promising technique to solve the inverse problem in imaging. The mathematical model of EIT reconstruction can be denoted as follows:

$$R_{\text{EIT}} = \arg\min_{R_\theta,\boldsymbol{\theta}} \sum_{n=1}^{N} \mathcal{L}\left(\boldsymbol{x}_n^{\text{Recon}}, \boldsymbol{y}_n^{\text{Label}}\right) + R(\boldsymbol{\theta}). \tag{1}$$

In (1), $\{\boldsymbol{x}_n^{\text{Recon}}, \boldsymbol{y}_n^{\text{Label}}\}_{n=1}^{N}$ are the training sample, $\arg\min_{R_\theta,\theta}$ is the deep learning architecture, $\mathcal{L}(\cdot)$ is the loss function, $R(\boldsymbol{\theta})$ is the regularized function. In this article, differential imaging is used, i.e., the difference between the measured voltage and the reference state at different moments is used to reconstruct an image of the conductivity change.

A. Method Overview

The overall framework of DHU-Net is presented in Fig. 1, which is modified by a U-shaped structure and mainly consists of three parts, i.e., the dual-branch encoder, fusion decoder, and attention-based skip connection. In particular, the encoder involves two branches for shape modeling with DCNs and channel-wise attention, conductivity representation with spatial attention, and HyperConvNets, respectively. The decoder has a similar symmetric structure, which is used to decode and expand the latent features learned by the encoder step by step to achieve image reconstruction and detail sharpness. Inspired by the improvement of U-Net and its novel models [52], DHU-Net adds skip connections between the corresponding feature encoder and decoder in a symmetric manner, which helps to recover fine-grained details in the prediction. Unlike ordinary skip connections, DHU-Net bridges the gap between the encoder and decoder by designing a new nonlocal network

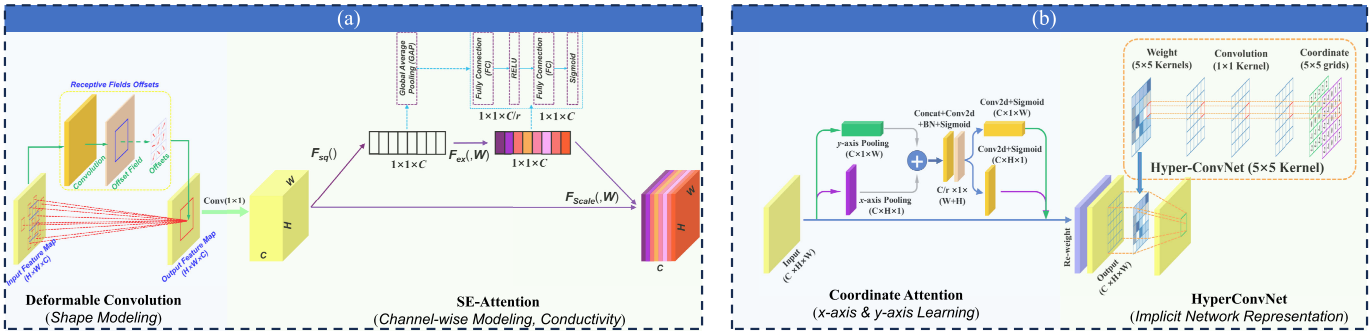


Fig. 2. Feature extraction module in DHU-Net. (a) DCN-SE attention to shape modeling. (b) CA-HCs for lightweight conductivity representation.

based on attention manner. In Sections II-B and II-C, the details of DHU-Net will be introduced, while some figures are also demonstrated.

B. Encoder

The input of DHU-Net is a 2-D matrix indicating the initial imaging result by numerical methods. The channel expansion is embedded before the encoder using an embedding layer as shown in Fig. 1, i.e., a 1×1 convolution is used to transform the input patch $\mathcal{X} \in \mathbb{R}^{1\times H\times W}$ into a high-dimensional tensor $\mathcal{X}_0 \in \mathbb{R}^{C\times H\times W}$. The Encoder of DHU-Net, as the core part of feature extraction and latent representation, designs a novel two-branch learning architecture, which uses a 3×3 local deformable convolution kernel and channel-wise attention for shape modeling, and CA with a 5×5 implicit hyper-convolution for global learning, respectively. This idea can effectively reduce the number of parameters of the model while improving the flexibility of the network structure.

1) DCN With SE-Attention Operator: DCNs are novel networks that introduce deformable sampling into traditional convolution. It allows the convolution kernel to sample locally on the input feature map, which enables the model to capture the details, shape, and spatial relationship of the target more accurately, and improves the modeling capability; second, DCNs extend the receptive field by sampling offsets and improves the model's perception of the global structure, and in addition, compared to regular convolution, DCNs do not require any additional parameters and achieves the deformation modeling by sampling point adjustments, which can improve the model's expressiveness while keeping the number of parameters and computation amount relatively small. The 3×3 deformable convolutional kernel with dilation 1 is utilized in DHU-Net for learning the shape of inclusions, i.e., the deformable convolution kernel can be expressed as

$$\mathcal{K} = \{(-1,-1),\ldots,(0,0),\ldots,(1,1)\}. \tag{2}$$

Let $\mathcal{X}_{\text{input}} \in \mathbb{R}^{C\times H\times W}$ and $\mathcal{X}_{\text{output}} \in \mathbb{R}^{C\times H\times W}$ denote the tensor of input and output, respectively, and the w_k, p_0, and Δp_k denote the learned weight, location of output map, the locations of enumeration in the kernel, and offsets (where $k = 1, 2, \ldots, |\mathcal{K}|$). The DCN can be modeled as

$$\mathcal{X}_{\text{output}}(p_0) = \sum_{p_k\in\mathcal{K}} w_k \cdot \mathcal{X}_{\text{input}}(p_0 + p_k + \Delta p_k) \tag{3}$$

where the offset Δp_k is usually a fraction, the operation process of (3) can be implemented by bilinear interpolation [30], which is shown in Fig. 2(a). In practice, a regular convolution layer is used to obtain a feature map with the same tensor as the input feature map, and the offset field has the same spatial resolution ($C \times H \times W$) as the input feature $\mathcal{X}_{\text{input}}$ and the output feature $\mathcal{X}_{\text{output}}$, and the number of channels of the offset field is $2N$ corresponding to the N x-axis hyper-parameters and N y-axis hyper-parameters, respectively.

After DCN for feature modeling of inclusions, the output feature maps focus more on the representation of the structure, however, ignore the global-level information. DHU-Net adaptively extracts the different weights between channels by introducing the squeeze-and-excitation Attention (SE-Attention). The SE channel-wise attention is composed of two main parts, as shown in Fig. 2(a), the first part uses global average pooling (GAP) to obtain global information $\mathcal{Z} = [z_1,\ldots,z_c,\ldots,z_C] \in \mathbb{R}^{C\times1\times1}$ on the different channels (denoted as $F_{\text{sq}}(\cdot)$) in figure). For a C channel feature map, the cth channel feature map by GAP can be obtained as follows:

$$z_c = \frac{1}{H\times W}\sum_{i=j=1}^{H,W} \mathcal{X}_{\text{input},c}(i,j),\ z_c \in \mathbb{R}^{1\times1}. \tag{4}$$

And then, the second part uses a multilayer perceptron (MLP) to nonlinearly transform the features to generate a channel attention tensor $\mathcal{S} = [s_1,\ldots,s_c,\ldots,s_C] \in \mathbb{R}^{C\times1\times1}$, which is normalized by a nonlinear function "sigmoid" that is used to adjust the weights of each channel (denoted as $F_{\text{ex}}(\cdot, W)$ in figure). Finally, the attention tensor $\mathcal{S}$ is used to allocate the weights of the input feature [as $F_{\text{scale}}(\cdot)$ in Fig. 2(a)], i.e.,

$$\mathcal{X}_{\text{output}} = \mathcal{S} \times \mathcal{X}_{\text{input}} \in \mathbb{R}^{C\times H\times W}. \tag{5}$$

The encoder demonstrated in (3)–(5) can be achieved shape modeling in local information while assigning different attention weights to global information, which enables this branch better focuses on key features and give reasonable channel-wise weights between input and output so that improve the feature discriminability.

2) CA With HyperConvNets: CA applies SA to coordinates in order to model the positional feature through spatial information so that the network can better perceive the correlation between different positions, thus better understanding the orientation-based features. The module is shown in Fig. 2(b). There are two main procedures in CA, i.e., the precise location is used to encode the long-range dependency in the feature modeling, where a pair of orthogonal 1-D pooling is utilized for coordinate embedding to aggregate the input tensors $\mathcal{X}_{\text{input}} \in \mathbb{R}^{C\times H\times W}$ along the horizontal (x-axis) and vertical (y-axis) orientations, respectively, to obtain a pair of coordinate-aware feature maps $\mathcal{Z}^h \in \mathbb{R}^{C\times H\times 1}$ and $\mathcal{Z}^v \in \mathbb{R}^{C\times 1\times W}$. This computation can be demonstrated as

$$\begin{cases} \mathcal{Z}^h = \left[z_1^h, \ldots, z_c^h, \ldots, z_C^h\right], & z_c^h = \dfrac{1}{W}\displaystyle\sum_{i=0}^{W} x_c(h, i) \\ \mathcal{Z}^v = \left[z_1^v, \ldots, z_c^v, \ldots, z_C^v\right], & z_c^v = \dfrac{1}{H}\displaystyle\sum_{j=0}^{W} x_c(j, W). \end{cases} \tag{6}$$

This coordinate feature aggregation represents long-range dependencies along the axis-wise orientations and preserves sharper model features on the orthogonal feature maps. Next, coordinate feature computation is used to construct the spatial attention tensor and acts on the input feature tensor. The orthogonal feature maps in (6) are augmented and concatenated, and channel adjustments are implemented using a 1×1 convolution and nonlinear projection to obtain the intermediate feature map $\mathcal{F} \in \mathbb{R}^{(C/r)\times 1\times(H+W)}$ (where r is the channel reduction ratio). The feature map $\mathcal{F}$ that encodes spatial information in both the horizontal direction and the vertical direction is split and the x- and y-axis features are encoded again using 1×1 convolution and nonlinear projection to obtain the features $\mathcal{G}^v \in \mathbb{R}^{C\times H\times 1}$ and $\mathcal{G}^h \in \mathbb{R}^{C\times 1\times W}$. Above all, the CA block is implemented as

$$\begin{cases} \mathcal{X}_{\text{CA}} = \left[x_{\text{CA}}^1, \ldots, x_{\text{CA}}^c, \ldots, x_{\text{CA}}^C\right] \\ \mathcal{X}_{\text{CA}} = \mathcal{X}_{\text{input}} \times \mathcal{G}^h \times \mathcal{G}^v \\ x_{\text{CA}}^c(i, j) = x_{\text{input}}^c(i, j) \times g^h(i, 1) \times g^v(1, j) \end{cases} \tag{7}$$

where, $\mathcal{X}_{\text{CA}}$ denotes the CA tensor, x_{CA}^c $(c = 1, 2, \ldots, C)$ denotes the cth coordinate feature map, $g^h(i, 1)$ and $g^v(1, j)$ denote the vectors of $\mathcal{G}^h$ and $\mathcal{G}^v$, respectively. When the CA feature map $\mathcal{X}_{\text{CA}}$ is obtained, the global feature is performed from the input feature map.

However, large-scale convolution kernels (e.g., 5×5, 7×7, etc.) will need large computational resources while the robustness of the model is ineffective. DHU-Net uses *hyperconvolution* with implicit feature learning to characterize the global parameter information of the 5×5 convolution operation. The core idea of hyperconvolution is an implicit representation of a convolution kernel, where the input coordinate maps with the same size as the convolution kernel and the use of a neural network to generate weights in the convolution kernel are implicitly coupled by a function representation of the hypernetwork. Unlike the standard convolution, the size of the convolution kernel is a design choice that does not affect the number of computational parameters, and the architecture of hypernet is shown in Fig. 2(b). Concretely, for a 2-D hyperconvolution kernel with 5×5 size, the input is the kernel spatial coordinates implicating the x- and y-axis

$$\{(i, j) | -2 \leqslant i \leqslant 2, \quad -2 \leqslant j \leqslant 2 \cup i, \ j \in \mathbb{N}\}. \tag{8}$$

The hyperconvolution is a function Φ_θ parametrized by θ, which maps the coordinate (i, j) to the corresponding weights of kernel $\mathcal{K}(i, j)$. The hyperconvolution can be mathematically modeled as

$$\mathcal{K}(i, j) = \Phi_\theta(i, j). \tag{9}$$

In DHU-Net, a fixed node full convolutional network (FCN) mapping (13) with four layers of 1×1 convolutional kernel is used. The number of nodes in the last layer is denoted by N_L as hyperparameter indicating the capacity of the HyperConvNet. There are two observations of hyperconvolution.

1) The method utilizes implicit kernel functions for hyper-convolutional computation and avoids the problem of a large number of trainable parameters and computation resources in standard convolution. The implicit kernels can be dynamically generated when needed, without explicitly storing all the kernel weights, which improves computational efficiency. In specific, the total number of trainable parameters of hyperconvolution with L layers can be calculated

$$(N_L + 1) \times C_{\text{input}} \times C_{\text{output}} + \sum_{l=0}^{L-1} (N_l + 1) \times N_{l+1} \tag{10}$$

where the channel of input and output features are C_{input} and C_{output}, N_L is the neuro numbers of lth layer, and N_L is the number of channels of the final hidden layer in the hyper-network. It can be seen through (10) that the number of computational parameters of hyperconvolution does not depend on the kernel size, but mainly on the number of input and output channels and the number of nodes N_L in each layer. For most CNNs, the number of input-, hidden features, and output-channels are often larger than 8, in contrast, the number of channels in the hidden layers in Φ_θ is set to be smaller than 8 $(N_l < 8)$. Considering these assumptions, the amount of the dominant parameter of hyper-convolution operator can be approximately expressed as $(N_L + 1) \times C_{\text{input}} \times C_{\text{output}}$, so that for a fixed number of parameters, the proposed method can realize a dense kernel with an arbitrarily large receptive field that captures the high-resolution contextual information in the features. Therefore, without increasing the number of trainable parameters, hyper-convolution can provide extended receptive fields with high capacity, thus limiting the risk of overfitting.
2) The robustness of hyper-convolution. It has been previous theoretically proved that implicit structures have a bias toward learning "low-frequency functions" of spectral bias [53], hyper-convolution correlates kernel weights during initialization and gradient-based updating, regularizing the convolutional kernel weights and making them spatially smoother. It is assumed that the smoothing kernel will suppress the high-frequency components of the input features, which is an advantage in the presence of noise, such as pixel-wise additive noise, or adversarial perturbations, so suggesting that hyper-convolution can play an important role in the robustness of the deep model.

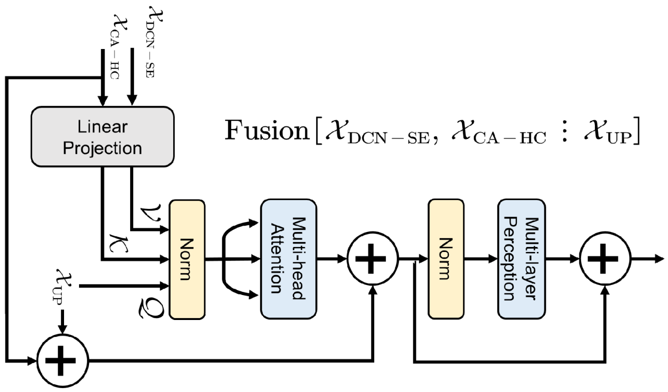


Fig. 3. Attention-based skip connection between encoder and decoder.

C. Decoder and Skip-Connection

The architecture of image reconstruction in the decoder is highly symmetrical to feature extraction in the encoders, where we use deconvolution to up-sample low-resolution features to high-resolution features, and are also merged by skip connections to capture both semantic and fine-grained information. Considering the dual-branch feature encoding structure of Encoder, this article designs a skip connection based on the SA to replace the feature concatenation in general U-shaped structural modules, whose architecture is shown in Fig. 3.

To be specific, the output of DCN with SE (DCN-SE) and CA with HyperConvNet (CA-HC) blocks, $\mathcal{X}_{\mathrm{DCN-SE}}$ and $\mathcal{X}_{\mathrm{CA-HC}}$, are transformed and split into a value matrix $\mathcal{V}$ and key matrix $\mathcal{K}$ after the linear projection (i.e., a one-layer neural network), Meanwhile the other input to the corresponding decoder is the up-sampled feature map $\mathcal{X}_{\mathrm{UP}}$ denoting as query matrix $\mathcal{Q}$, as shown in Fig. 3. Then, the SA computation is conducted in the decoder like

$$\mathrm{SA}(\mathcal{Q}, \mathcal{K}, \mathcal{V}) = \operatorname{soft\,max}\left(\frac{\mathcal{Q} \cdot \mathcal{K}^{\mathrm{T}}}{\sqrt{d_k}}\right) \times \mathcal{V} \tag{11}$$

where d_k denotes the scale of the flatten vector. After feature fusion in SA, the decoder module uses two 3×3 convolution operations for channel-wise feature aggregation and feature recovering, the output is used as input features for the next decoder module.

D. Loss Function

The loss function is defined as

$$\begin{aligned} \mathrm{Loss} &= \mathcal{L}_{\mathrm{MSE}} + \lambda \cdot \mathcal{R}(\boldsymbol{\theta}) \\ &= \sum_{(\hat{x}, y) \in \mathcal{D}} \left\| \hat{x} - f(y|\boldsymbol{\theta}) \right\|_2^2 + \lambda \cdot \|\boldsymbol{\theta}\|_2^2 \end{aligned} \tag{12}$$

where, $\hat{x}$ is the ground-truth data as the supervised label, y is the initial reconstruction results as the DHU-Net input, $\mathcal{D}$ is the given training samples, f denotes the network output under the inputs and hyper-parameters $\boldsymbol{\theta}$ (such as the kernel weights, coordinate maps, and so on). The first term $\mathcal{L}_{\mathrm{MSE}}$ is used to ensure that the reconstruction is as close as possible to the ground-truth, while the second term is regularization term based on $L2$-norm penalty function avoiding overfitting problems during training, and λ is the coefficient setting as 0.01 in DHU-Net. The process of network training is to find the optimal network parameters $\boldsymbol{\theta}^*$ to enable the network to image satisfactory reconstruction $\boldsymbol{y}^*$.

III. Experimental Settings

This section gives the experimental details of DHU-Net, including the dataset collection used for model training, the initialization method, the parameter configuration, comparison methods, and gives quantitative metrics used to evaluate the experimental performance.

A. Datasets

The datasets for DHU-Net training and validation are simulations based on numerical calculations, and the information related to building the simulation model, computation of the forward problem, and discrete representation of the conductivity for the inverse problem is given in Fig. 4.

As shown in Fig. 4(a), a circular cross section is established an electrical sensitive field, and 16 electrodes are placed at equal intervals as excitation-measurement sensors. The adjacent excitation-adjacent measurement strategy with 4.5 mA amplitude and 100 kHz excitation frequency current is used for collecting the response. The diameter of the observation domain is set as 0.19 m while the lengths of the inclusions are set as in the range of 0.02–0.08 m, the conductivity of water as the background is set as 0.06 S/m, and the medium is simulated using different conductivities set as a random value in the range of 10^{-6}–10^{-12} S/m. The inclusions with random position are placed in dataset, and they do not intersect each other. Overall, this dataset includes 42 430 circular samples and 400 samples of other shapes, a total of 42 830 simulation training samples are included, of which the setup training samples, and test samples each account for 85%, and 15% of the total database (where the training samples, test samples are not duplicated but contain similar shape inclusions and distributions).

The forward problem of EIT based on the observation model (4) is solved by finite elements method (FEM). With the help of the COMSOL Multiphysics with MATLAB, the observation domain is discretized into a triangular mesh and then solved, as shown in Fig. 4(b). For the 16-electrode EIT measurement model shown in Fig. 4(a), a total of 208 voltages are obtained for all electrodes excitation-measurement at one time [23], [54], [55]; in addition, the relationship between the conductivity distribution and the boundary measurement voltage is obtained by calculating the Jacobian matrix [29]. The sensitivity matrix for the case of 1–2 electrode excitation and 8–9 electrode measurement is represented as shown in Fig. 4(c). The differential imaging is used in DHU-Net to reduce the perturbation in the measurement system. The measurement without any inclusions in the observation area is set as the reference voltage [as shown in Fig. 4(d)], the measurement when there are different distributions with various phases is the measurement voltage [as shown in Fig. 4(e)], and the difference between the two is used as the input of DHU-Net.

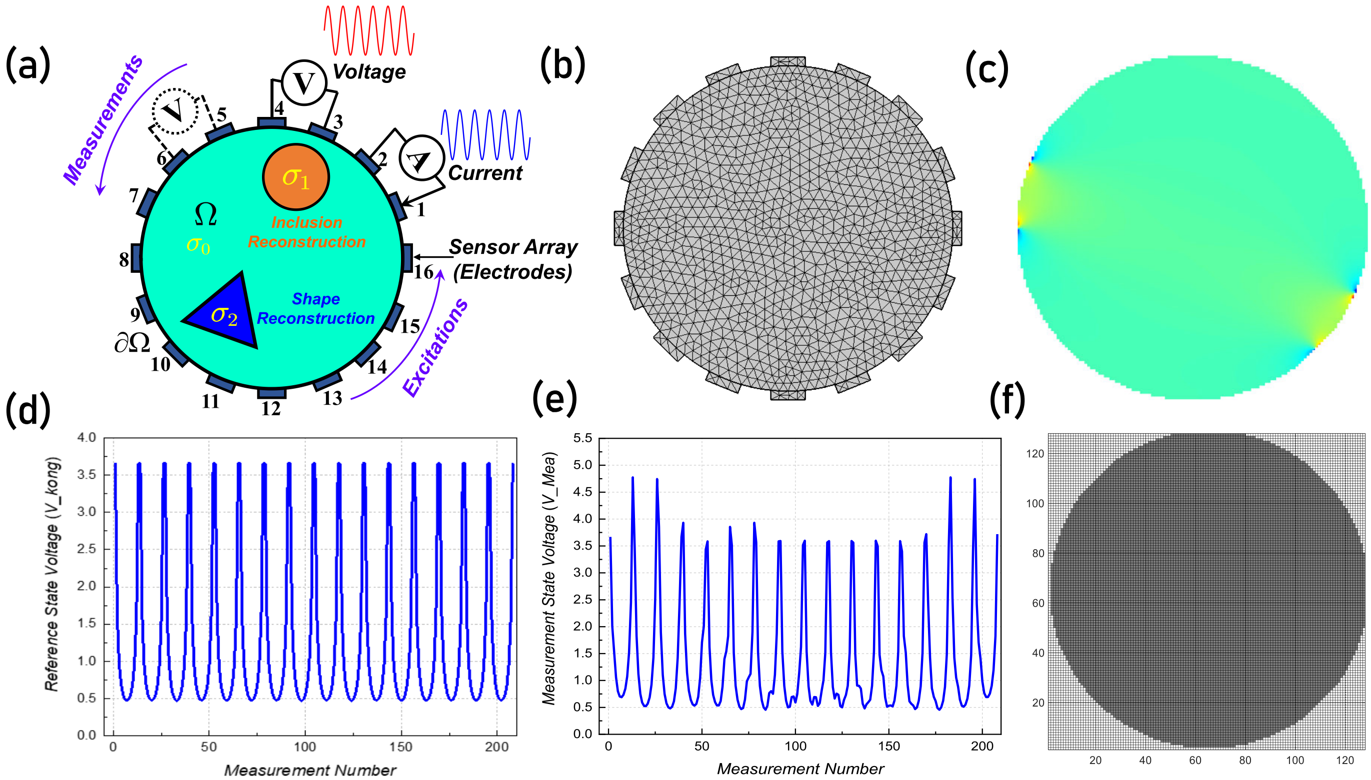


Fig. 4. Information of simulated EIT datasets. (a) EIT Sensor model with 16-electrodes for image reconstruction and shape reconstruction. (b) FEM mesh for solving forward problem. (c) Sensitivity map (1–2 electrodes excitation and 8–9 electrodes measurement). (d) and (e) Measurement voltage under homogeneous scenario (without inclusions) and inhomogeneous scenario (with inclusions). (f) Grids for inverse problems representing the discrete conductivity distributions.

In order to avoid the problem of "inverse crime," different discrete meshes are used to characterize the conductivity distribution in the domain [56]. The observation domain is divided using square grids into 128 × 128 pixels to adequately represent information about inclusions with complex boundary shapes. The conductivity of the background with 0.6 S/m is set as 1, and the other pixels are normed between [0, 1] using a linear normalization method.

Each sample in the datasets contains two independent vectors, i.e., the differential voltage vector $U \in \mathbb{R}^{208\times 1}$ and the discrete conductivity matrix $x \in \mathbb{R}^{128\times 128}$ corresponding to the voltage. In addition, the sensitivity map is utilized for manifold transfer from the measurement voltage to the "coarse" distribution. The voltage is used as input for forward propagation to obtain the reconstructions, and the label and predicted media distribution are used as independent variables of the loss function to supervise the training process of the deep learning model.

B. Initialization

1) Manifold Transfer: The learning-based methods can benefit from warm starting, i.e., initialization near a suitable solution for the objective function. For ill-posed problems, the solution with the $L2$-norm operator is adopted to initialize the DHU-Net

$$x_{\text{initial}} = \left(J^{\mathrm{T}}J + \alpha L^{T}L\right)^{-1} J^{\mathrm{T}}U. \tag{13}$$

Equation (13) transfers the voltage to the initial conductivity distributions, where the L is regularization pattern, $\alpha = 0.01$ is the coefficient.

2) Hyperparameter Initialization: The mini-batch is utilized to train the DHU-Net with a size 128 and the total training steps are 400, and the optimizer is the Adamax with parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$. The updating learning rate used in all CNN-based experiments with an initial learning rate of 0.1 and

$$\eta_t = \eta_{t-1} \cdot \left(1 - \frac{t}{T}\right)^{0.9} \tag{14}$$

where the η_t, η_{t-1}denote the tth and (t-1)th learning rate, t denotes the current training step and T denotes the total training step, and $\varepsilon = 10^{-8}$ is the indicator for stopping the training. The weight parameters are randomly set to Gaussian-distributed random numbers with mean 0 and standard deviation 0.01. The model was performed on a PC (Windows 10 + Intel Core i7-9700K @ 3.6 GHz CPU RAM 32 GB + NVIDIA 2080Ti GPU) in Python 3.8, TensorFlow 1.14.0, and Keras package environments, with a total of 400 steps of model training in about 38 h.

C. Model Configuration

The typical U-Net is utilized as the backbone in DHU-Net, and a two-branch encoder is used to extract local and global features, where the dimensions (referring to the height and the width) of the feature maps are 128, 64, 32, 16, and 8, and the corresponding channels are 8, 16, 32, 64, and 128, respectively. In the DCN-SE local representation branch, the channel reduction ratio of the FC layer is set to $r_{\mathrm{SE}} = 4$, 4, 8, 8, 16, and in the CA-HC global representation branch, the channel reduction ratio of the FC layer is set to $r_{\mathrm{CA}} = 4$, 4, 8, 8, 16. The node for the hyperconvolution is set to $N_L = 4$. The two adjacent modules of the encoder are connected using average pooling with the pooling kernel size set to 2×2. Symmetrically, up-sampling in the decoder uses deconvolutional, with a convolutional kernel dilation of 3.

D. Comparisons

To assess the performance of the DHU-Net reconstruction proposed in this article, we use the numerical algorithm, as well as some advanced EIT image reconstruction methods, including CNN-based and Attention-based frameworks.

1) Numerical Algorithm: The TR [57] was chosen as the numerical method due to the fact that the TR algorithm quickly approaches the optimal solution by a simple regularization pattern.

2) Learning-Based Methods: The vanilla U-Net is adopted as the backbone model, and two broad approaches are involved in comparative experiments, i.e., CNNs-based and Attention-based imaging methods, including VD-Net [58], VA-Net [29], HIHU-Net [59], and HybridDenseU-Net [60]. VD-Net and VA-Net are two new deep imaging frameworks that incorporate dense connection and attention with a U-shaped structure as the backbone, HIHU-Net is a multiscale dense connection encoding-decoding model for EIT image reconstruction, and HybridDenseU-Net is a dual-branching encoder, feature fusion decoder-style symmetric image reconstruction framework.

E. Evaluation Metrics

For a quantitative evaluation, the root mean square error (RMSE) and structural similarity index (SSIM) are measured as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\|x_{\mathrm{Rec}} - x_{\mathrm{label}}\|_2^2} \tag{15}$$

$$\mathrm{PSNR} = 20\log_{10}\frac{\max(x_{\mathrm{Rec}})\sqrt{N}}{\|x_{\mathrm{Rec}} - x_{\mathrm{label}}\|_2} \tag{16}$$

$$\mathrm{SSIM} = \frac{(2\mu_{\mathrm{Rec}}\mu_{\mathrm{label}} + C_1)(2\Sigma_{\mathrm{Rec,label}} + C_2)}{(\mu_{\mathrm{Rec}}^2 + \mu_{\mathrm{label}}^2 + C_1)(\Sigma_{\mathrm{Rec}}^2 + \Sigma_{\mathrm{label}}^2 + C_2)}. \tag{17}$$

The Rec denotes the reconstructions, label denotes the ground-truth distributions, N denotes the pixels in the image, and μ, Σ are the mean and covariance matrix, respectively. The RMSE and PSNR evaluate the pixel-wise between reconstructions and ground-truth, and SSIM measures the similarity of the feature contents between reconstructions and references, where lower RMSE and higher PSNR mean better results, while larger SSIM corresponds to a more competitive result.

IV. Results

To illustrate the efficacy of the DHU-Net proposed in this article in nonlinear soft-field EIT imaging, a series of simulation experiments and physical experiments have been conducted, and the related visualization results and quantitative metrics analyses are given in detail below.

A. Numerical Results

We randomly selected test samples for visualization experiments and compared the reconstruction results of relevant numerical solution methods and deep learning-based methods, the results are presented in Fig. 5(a). Among them, models 1 and 2 are the inclusions reconstruction, models 3 and 4 are the shape reconstruction, the first column is the model set, and the second to seventh columns (from left to right) are the reconstruction results using different imaging methods, respectively; the corresponding quantitative metrics results (RMSE, SSIM, and PSNR) are shown in Fig. 5(b). We performed the quantitative metrics calculations using the simulation test samples and chose the average values as the results for analysis.

As shown in Fig. 5(a), it can be seen that the deep learning-based reconstruction performs better than the regularization method (TR), and it can be seen that there are no obvious artifacts with the reconstructed image and the conductivity parameters and boundary shape information can be seen more clearly. Comparing the results of the deep learning-based imaging methods (VD-Net, VA-Net, HIHU-Net, HybridDenseU-Net, and DHU-Net proposed in this article), we can see that VD-Net, VA-Net, and HIHU-Net use a single-scale encoder for feature extraction, and although the result interference caused by the artifacts can be improved to some extent, the reconstruction results (especially in the shape reconstruction task) show obvious distortion and poor shape representation performance. Also, for example, targets with small scales, in the Model 2 samples, do not appear reconstructed in the VD-Net results, suggesting some limitations of simple feature aggregation ideas. The HybridDenseU-Net and DHU-Net use two-branch encoder and perform feature fusion, the reconstruction results are significantly improved and the "sharp" features are well preserved, a further comparison between the two methods reveals that DHU-Net achieves more optimal results due to the use of deformable convolution and CA for shape information.

The quantitative metrics shown in Fig. 5(b) have results consistent with the above analysis. Fig. 5(b) analyses the performance between both RMSE and SSIM on different algorithms, and it can be seen that DHU-Net has the optimal RMSE and SSIM results (i.e., the lowest RMSE is 2.028 and the optimal SSIM is 0.9631 among the comparison methods used), which indicates that the reconstruction results and the real distribution of DHU-Net are in good agreement. In addition, the PSNR metrics are quantified for the reconstruction results of all the simulation test data, as shown in Fig. 5(b), it can be seen that the reconstruction of the deep imaging methods using single-scale encoder (VD-Net, VA-Net, and HIHU-Net) are unstable, which can be affected by the distribution of the model, and the complexity of the shapes, whereas the multiscale Encoders, HybridDenseU-Net, and DHU-Net methods, have better PSNR results, and the results of the DHU-Net are in better consistency

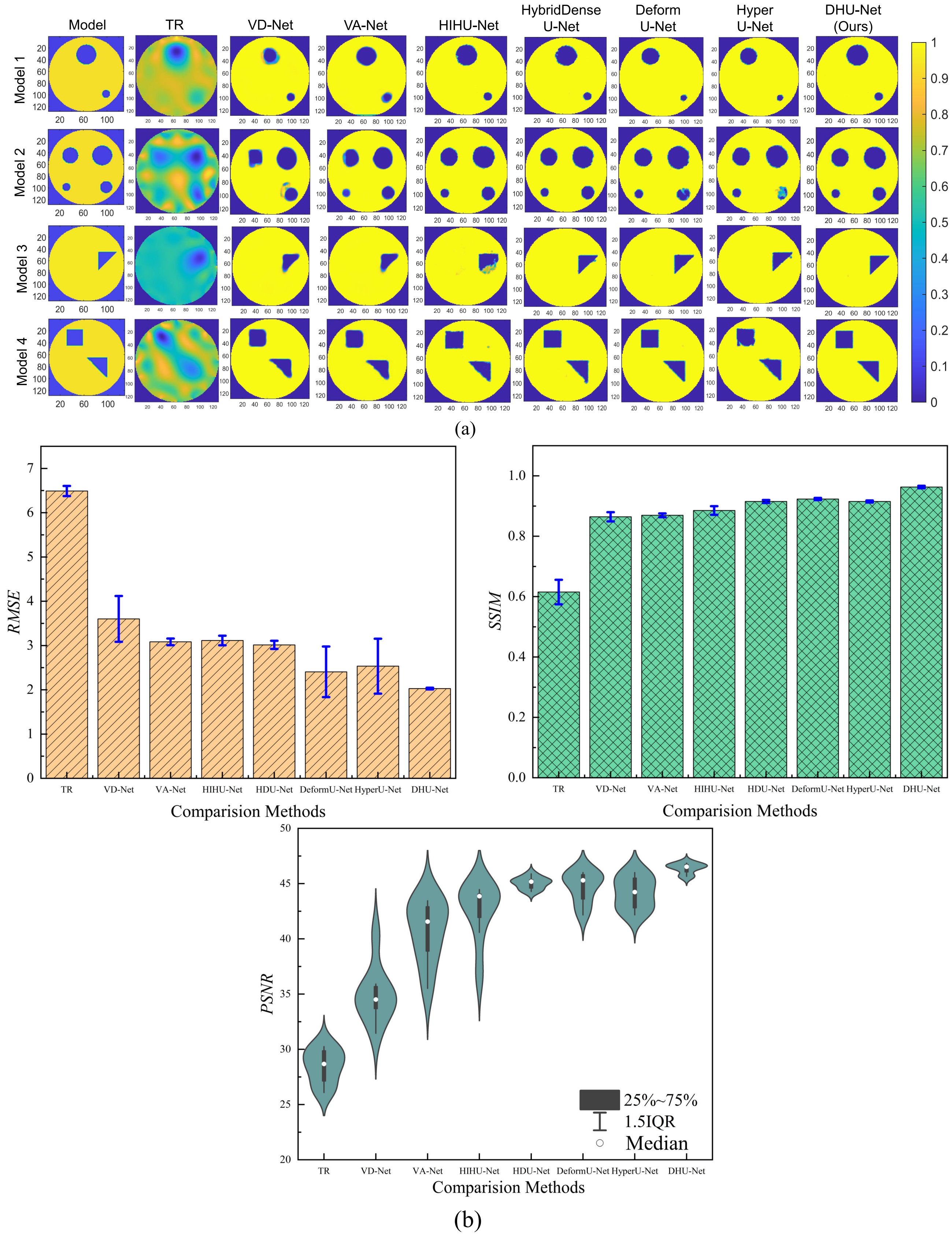


Fig. 5. Reconstruction results of the different methods under simulation models. (a) From left to right, the ground truth and the reconstruction results of these methods. All the conductivities are normalized and displayed in ranges [0, 1]. The models 1 and 2 show the inclusion reconstruction, and models 3 and 4 show the shape reconstruction. (b) RMSE, SSIM, and PSNR metrics in test datasets.

(i.e., there is not much variability of the results from the 25%–75% results of the quantified results).

B. Experimental Results

1) Experiments Setting: The experimental platform was acquired using the developed 16-electrode EIT data acquisition system. The parameter setting is consistent with the simulation environment: a 0.19 m diameter tank is used, the homogeneous background (reference measurement state) is simulated using an NaCl solution with a conductivity of 0.06 S/m, and the inhomogeneous phase (different media distribution) is simulated using glass rods (taking into account that the glass is insulating, the conductivity of the inclusions is assumed to be 10^{-12} S/m). The EIT-DAS has a signal-to-noise ratio (SNR) of 70 dB, a bit resolution of 14 bits, an excitation current with an amplitude magnitude of 4.5 mA, and excitation frequency is 100 kHz, and consists of a sensing array, an FPGA master, a digital/analog converter, a voltage-controlled-current-source (VCCS), a programmable variable gain amplifier (PGA), and a PC for results visualization.

2) Experiment Result: As shown in Fig. 6(a) and (b), the deep learning-based imaging methods have better reconstruction quality than the TR methods (consistent conclusions in both visualization and quantitative metrics), while VD-Net, VA-Net, and HIHU-Net do not obtain satisfactory results (the "sharp" parts of the reconstructions are obviously smoothed and some small objects are presented incorrectly, etc.) in reconstructing small-sized objects and complex shapes of inclusions, as they only consider the coding of spatial and parametric information at one scale. The corresponding quantitative metrics are also consistent in their conclusions. Although both HybridDenseU-Net and DHU-Net use different convolutional kernels for feature coding, DHU-Net uses deformable convolutional kernel for shape modeling and an implicit neural network for latent feature construction to achieve good results in both reconstructed images and quantitative metrics.

V. Discussion

A. Robustness Analysis (With Respect to the Different SNRs)

In practical scenarios, fluid perturbations, circuit noise, and disturbances in the observation area can cause some perturbations to the measurement signals, so the anti-noise performance of the imaging method is very essential. In order to test the robustness of DHU-Net to noise, we performed experiments with SNRs of Gaussian white noise in our test samples ranging from 90 to 20 dB, and evaluated the image quality using three quantitative metrics, RMSE, SSIM, and PSNR, respectively. The metrics analysis is shown in Fig. 7. (The visual reconstructions are shown in the supplementary materials).

It can be seen that DHU-Net has good noise robustness. The quantitative results in Fig. 7 compare the DHU-Net with the VA-Net and HybridDenseU-Net for anti-noise testing, respectively. The results show that the strategy of multiscale feature coding and latent feature fusion can improve the noise robustness of the deep imaging model. (That is, the RMSE, SSIM, and PSNR metrics of DHU-Net and HybridDenseU-Net outperform the results of VA-Net under different noise-levels.) Overall, the deformable convolution can improve the shape modeling capability, and the implicit representation of hyperconvolution can better filter out the high-frequency features (i.e., the noise component) and thus smoothly retain the dominant features in the image. In the metrics analysis in Fig. 7, the RMSE of the DHU-Net method proposed in this article stays between 2.00 and 2.15 during the gradual decrease of SNR, the SSIM stays above 0.95, which indicates that the DHU-Net's has good noise robustness.

B. Complex Reconstructions (With Respect to the Multiphase and Other Shapes)

The advantage of EIT is visualizing the distribution of conductivities inside the measurement domain, and thus the need to evaluate the reconstruction method in scenarios containing different conductivities (in particular, media distribution scenarios containing low contrast conductivities). We constructed three sets for multiphase distributions, and the TR method, HybridDenseU-Net, and DHU-Net are used to visualize the reconstruction results. Moreover, we selected the local conductivity parameters for the analysis (as shown by the red dashed line in the figure), and the reconstruction results and analyses are shown in Fig. 8. (The visual reconstructions are shown in the supplementary materials.)

The conductivity contrast is defined as

$$\delta_\sigma = \frac{\sigma_{\text{inclusion}}}{\sigma_{\text{background}}} \times 100\%. \tag{18}$$

As the results shown in Fig. 8: in the reconstruction results using the deep learning-based method, the imaging results have good performance when the inclusions have a large distribution and high-conductivity contrast, which can more accurately express the electrical distributions of the medium and clearly reconstruct the boundaries of the inclusions. The TR is unable to display information about inclusions with low-conductivity contrast when the conductivity contrast is gradually reduced, however, the HybridDenseU-Net and DHU-Net methods improve the imaging results to a certain extent and the approximate distribution of inclusions can be seen in the scene (but the boundaries have some distortions). As DHU-Net improves the robustness and expressiveness of the model by introducing the shape modeling method of deformable convolution and the parameter mapping of implicit hyperconvolution, the conductivity parameters are more accurate and the shape features are more clearly defined (the HybridDenseU-Net with conventional convolutional kernel feature expression still suffers from some artifacts/perturbations in the multiphase conductivity reconstruction task).

Fig. 8 analyzes the conductivity distribution in the ROI region. It can be seen that the sensitivity matrix-based method causes the results to be smoothed and the boundary features to be distorted due to the approximation of the higher-order nonlinearities. the HybridDenseU-Net reconstructed results have a high consistency with the references in the high-conductivity contrast, but when the inclusions are not obviously different from the background or in the multiphase conductivity model,

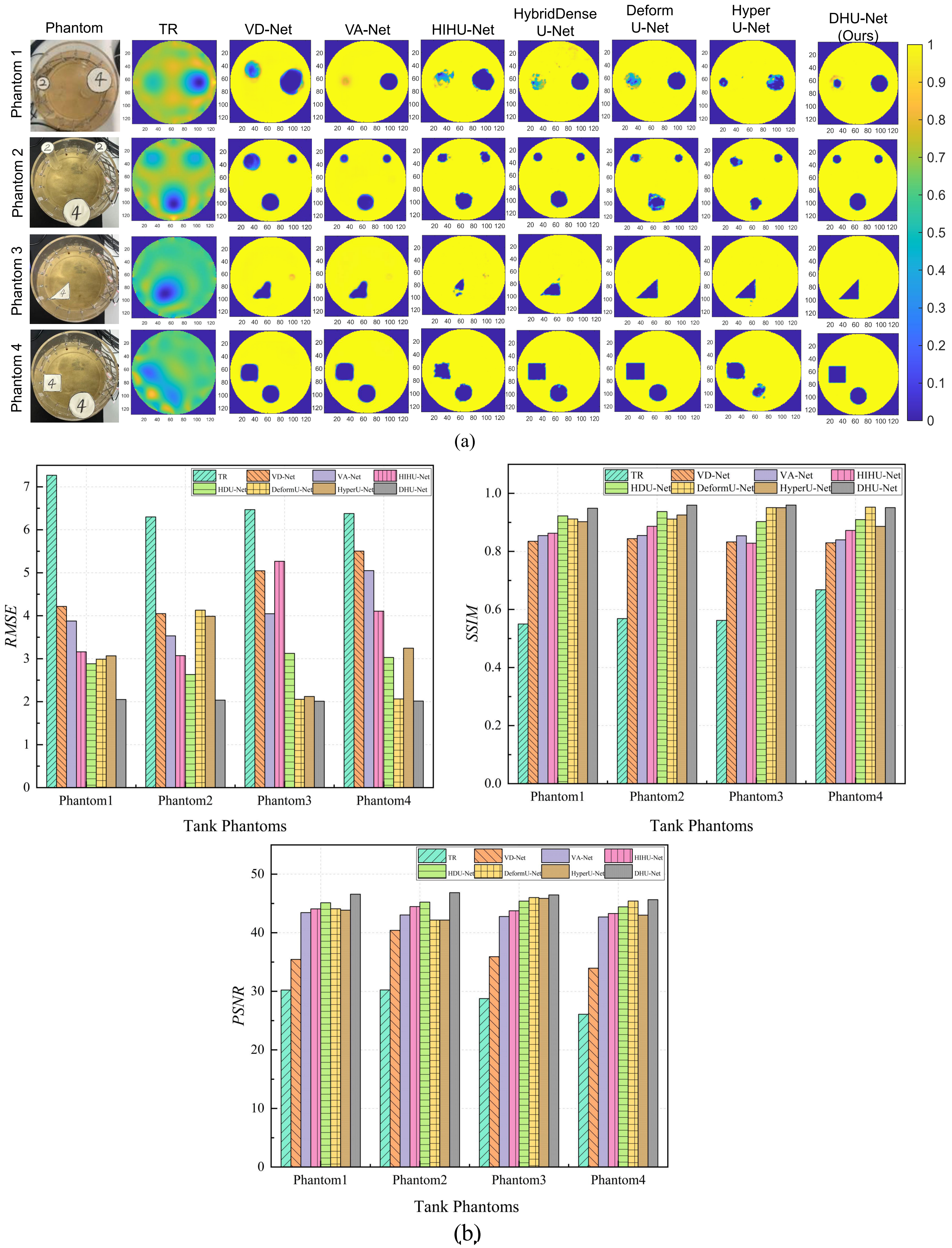
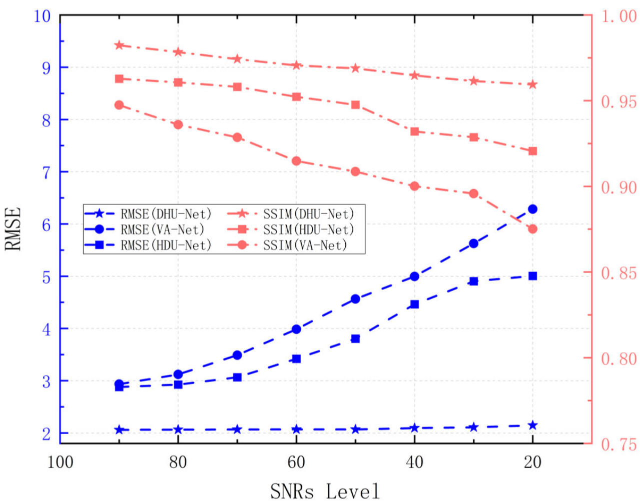
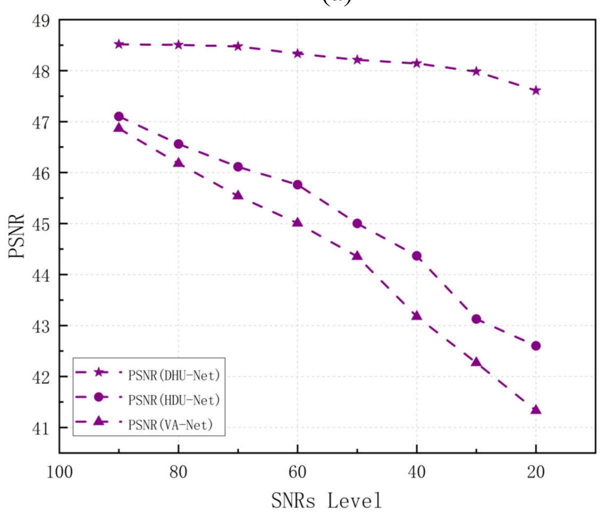


Fig. 6. Reconstruction results of the different methods under real-world tank experiments. (a) From left to right, the ground truth and the reconstruction results of these methods. All the conductivities are normalized and displayed in ranges [0, 1]. Phantoms 1 and 2 show the inclusion reconstruction, and phantoms 3 and 4 show the shape reconstruction. (b) RMSE, SSIM, and PSNR metrics.



(a)



(b)

Fig. 7. Quantitative metrics (RMSE, SSIM, and PSNR) of imaging results at different SNR levels. (a) RMSE and SSIM metrics with different SNR levels. (b) PSNR metric with different SNR levels.

the distribution of the conductivities can be observed to be in some error. DHU-Net has better generalization and robustness, while can reconstruct the position, shape, and electrical information of the distributions with multiphase conductivities (although there are errors in some regions, the imaging quality can be seen to be optimized significantly in the visualization results).

C. Ablation Study and Computational Resource

In this section, we discuss different feature connection, such as residual connection, dense connection, attention, and as used in this article deformable convolution and hyperconvolution to quantitatively analyze the imaging results (selecting RMSE and SSIM for quantitative analysis) and the number of model parameters. For a fair comparison, we both use the same training dataset and test dataset for quantitative analysis, with an imaging spatial resolution of 128×128, vanilla V-Net (3×3 convolutional kernel) selected for the baseline model, and the same training environment. The results are shown in Fig. 9.

1) Ablation Studies: From the reconstruction performance of the 11 models above [as shown in Fig. 9(a)], it can be seen that simple single-scale deep CNNs models (including the introduction of residual connection, dense connection, or attentions) have RMSE higher than 3.50 and the highest value of SSIM is 0.87 (HIHU-Net results), whereas multiscale feature encoding and latent feature fusion CNNs models have better reconstruction results (RMSE values below 3.00 and SSIM values above 0.90). It is shown that the feature encoder can learn different levels of features in the EIT reconstruction task at different scales and act simultaneously on the feature decoder. In addition, HybridDenseU-Net, HyperU-Net, and DeformU-Net use dense connection, implicit hyperconvolution, and deformable convolution alone, respectively, and it can be seen that the use of deformable convolution in different modules alone optimizes the quality of the reconstructed image, suggesting that deformable convolution operators play the role of shape modeling in the used for the graphic reconstruction EIT task, which effectively improves the model's expressive ability. Learning-based models using both deformable convolution and hyperconvolution have the lowest RMSE and optimal SSIM results, both acting simultaneously in CNNs models to improve the shape representation of the model by powerful shape modeling on the one hand and on the other hand, the smoothing regularity of hyperconvolution filters out the high-frequency noise and models global parameter distributions (conductivity distributions).

2) Computational Resources (Parameters and Computational Complexity): The analysis of Fig. 9(b) shows that the baseline model V-Net, due to the simple operations (only a fixed 3×3 convolution is utilized), has the largest RMSE and the lowest SSIM of the imaging results despite the smallest number of parameters and computational complexity, indicating that the simple model does not obtain a better result in the soft-field imaging problem. The number of Params and GFLOPs of the single-scale backbone based on dense connection is smaller than that of the two-branch model, while comparing the two-branch model with the single-branch backbone shows that the use of convolutional kernels of different scales for learning local features and global information improves the quality of the imaging, but it also increases the parameters of the model and the computational complexity. In contrast, the DHU-Net proposed in this article, although using a two-branch model, does not show a large increase in the number of parameters in the large-scale convolutional kernel operation, and the optimal RMSE and SSIM results simultaneously, because the implicit hyperconvolutional operator learns the convolutional kernel hyperparameters by inputting coordinate information which is not using so large amount learnable parameters.

3) Training Time and Imaging Time: In this part, the average imaging time of a section with different imaging methods in test datasets and the training time of deep learning methods are shown in Table I.

Considering both training time and testing time in Table I, it can be seen that although DHU-Net does not have the fastest imaging time, the DHU-Net proposed in this article can satisfy the demand for real-time imaging and the condition

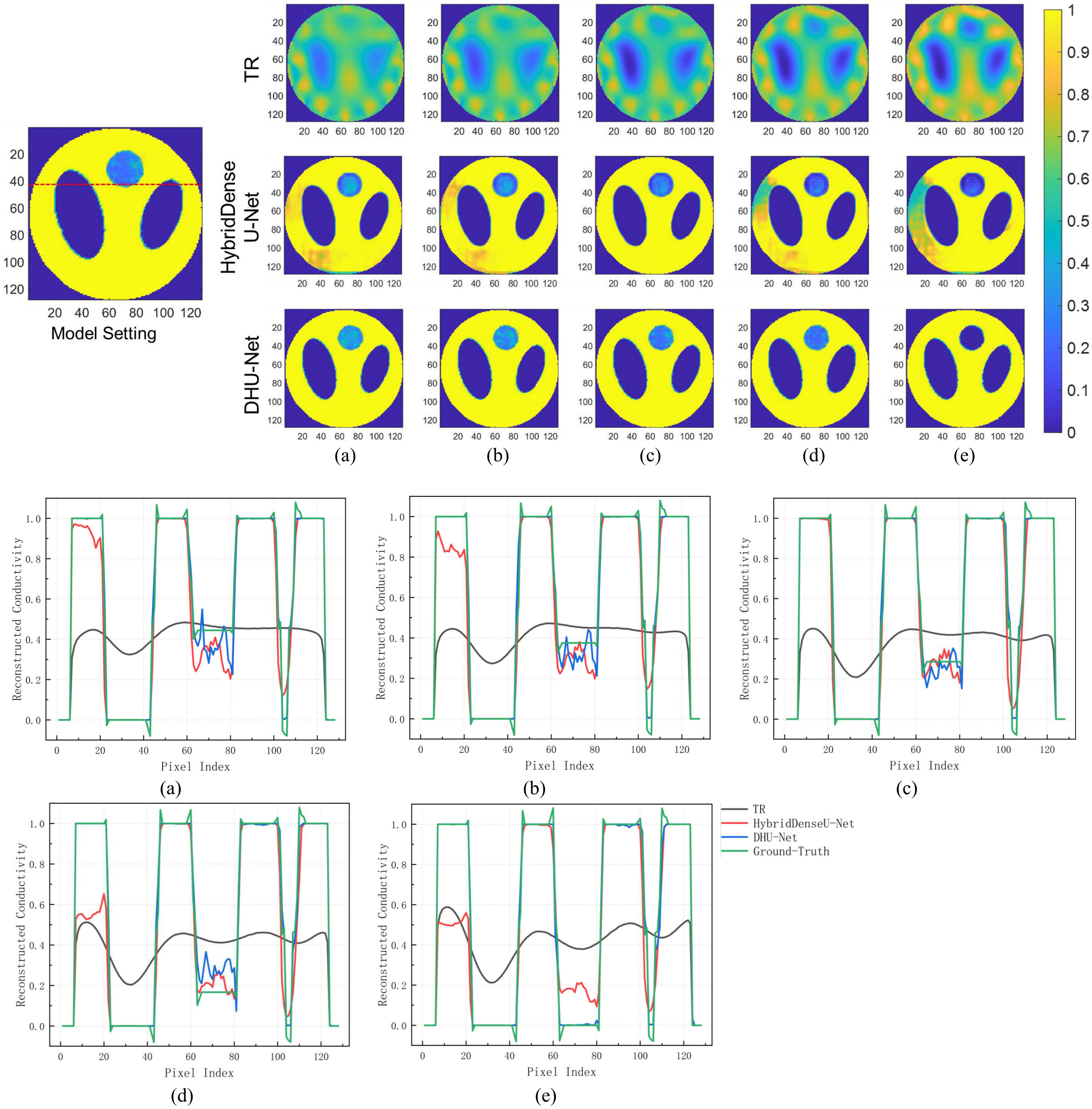


Fig. 8. Reconstructions with different conductivity contrasts (red dashed line indicates the area where ROI conductivity values were analyzed; results are shown at the bottom of Fig. 8). (Note: In this set of experiments, the parameters are set to fix the background conductivity and gradually decrease the high and low contrast values of the conductivity.) The reconstructions and corresponding distributions plots under different conductivities level, (a)–(e) 0.2×, 0.4×, 0.6×, 0.8×, and 1.0× lung conductivity.

of imaging consistency. If online imaging is performed on a high-performance computing platform, DHU-Net is expected to meet the requirements of dynamic imaging.

In summary, the DHU-Net proposed in this article improves the robustness and expressiveness of the model by introducing deformable convolutional and hyperconvolutional operators on the one hand, although the cost of a bit growth in the parameters and computational complexity that need to be trained (the analysis of this data was measured on reconstructed images with a pixel resolution of 128 × 128).

D. Limitations of the Proposed Methods

Although the DHU-Net method obtains superior results in shape reconstruction and multiphase conductivity reconstruction, it still has the following limitations.

1) As analyzed in Fig. 9 and Table I, model parameters and the imaging time of DHU-Net are not optimal, and further adoption of lightweight models for real-time dynamic imaging is needed for further improvement.
2) DHU-Net uses L_2 regularization information as input features, which may lead to errors in imaging results

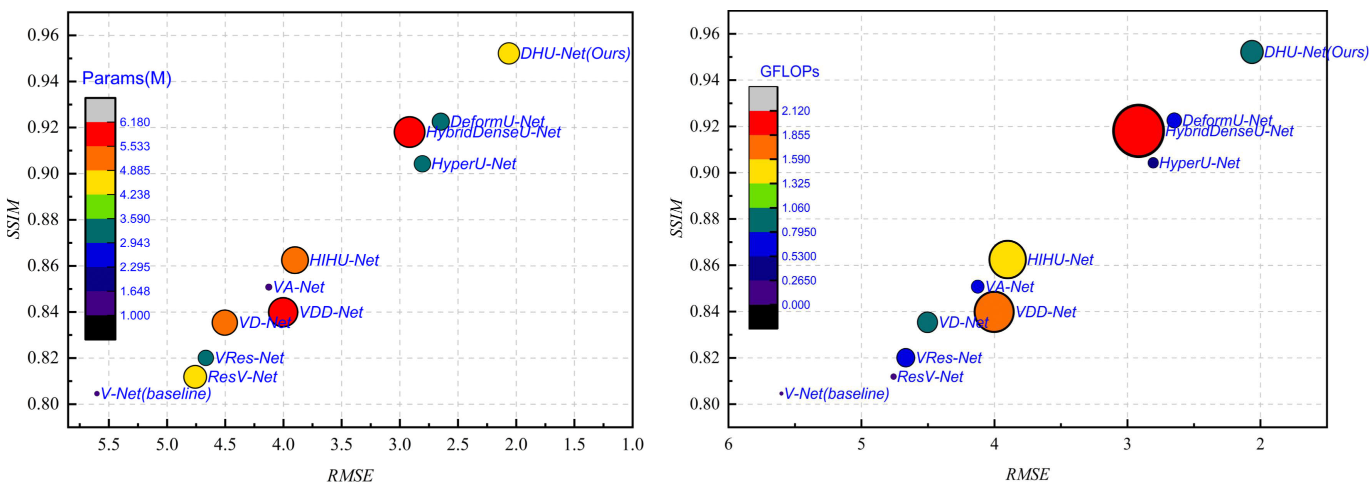


Fig. 9. Results of the ablation experiments (RMSE and SSIM) are analyzed with respect to the consumption of computational resources (number of model covariates) (where the baseline model is V-Net, the residual connectivity model is of class "Res," the dense connectivity model is of class "D," and the attention mechanism model is of class "A"). (a) Params with respect to the metrics (RMSE and SSIM). (b) GFLOPs with respect to the metrics (RMSE and SSIM).

TABLE I
TRAINING TIME AND INFERENCE TIME IN DIFFERENT METHODS

Models		Train (h)	Inference (s)
Comparison Models	V-Net(baseline)	**20**	**1.60**
	ResV-Net	54	3.30
	VRes-Net	36	3.28
	VD-Net	40	3.60
	VDD-Net	54	5.90
	HIHU-Net	50	4.20
	VA-Net	26	3.20
	HybridDenseU-Net	**55**	**6.14**
Ablation Studies	DeformU-Net	29	3.20
	HyperU-Net	26	2.60
	DHU-Net(Ours)	38	3.30

due to the loss of some "sharp" features. Therefore, some methods based on shape modeling [61] as the idea of implicit priori information, such as the level set method [62], [63], Fourier shape description [64], and other methods, combined with the topological evolution of deformable convolution, can express the shape information more flexibly.

VI. CONCLUSIONS AND OUTLOOKS

In this article, deformable convolution with better shape modeling ability and more robust hyperconvolution operator DHU-Net with multiscale feature encoder is proposed, and simulation experiments, water tank experiments, and multiple complex scene experiments are carried out using EIT in the nonlinear imaging inverse problem as an example. The experimental results show that the shape characteristics of sharp inclusions are better represented in the DHU-Net reconstruction results, which accurately reconstruct the multiphase distribution scenarios, while noise robustness experiments show satisfactory robustness at lower SNR (i.e., the measured signals contain a higher degree of perturbation). Although parametric analysis and ablation studies have shown that DHU-Net has a significant parametric increase, the method proposed in this article has better RMSE and SSIM compared to densely connected CNN-based models. The above analysis shows that the better structural and functional reconstruction performance of DHU-Net is expected to promote the application of EIT in the field of practical functional NDT such as industry and medicine. Meanwhile, further research on unsupervised/semi-supervised methods using models with fewer parameters, or employing other a priori knowledge fusion, is a direction in the field of imaging inverse problems.

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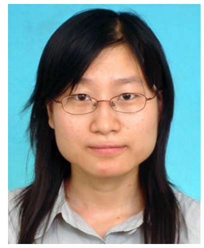

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